STATISTICAL QUALITY CONTROL · SIX SIGMA

# Analysis of Customer Waiting Time in Bank Service Queue

A process capability and control chart study applying SQC methods to identify and reduce service delays in banking operations.

SUBJECT
Statistical Quality Control & Six Sigma (6A03SQC)

PROGRAMME
Bachelor of Business Administration — Semester VI

DATA SOURCE
Kaggle Open Dataset

METHODOLOGY
Control Charts · Cp/Cpk · Pareto · Fishbone

Power BI | Power Query | Excel | X-R Control Charts | Process Capability

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2025

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# Statistical Quality Control Project Report

# Analysis of Customer Waiting Time in Bank Service Queue

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## 1. Introduction and Description of the Real-Life Process

In the service sector, quality is not only defined by accuracy but also by the speed and consistency with which services are delivered. In banking operations, customer waiting time has emerged as a critical quality parameter because it directly influences customer satisfaction, perceived service quality, and overall operational efficiency. Long and unpredictable waiting times often result in customer dissatisfaction, increased complaints, and a negative image of the bank. Therefore, controlling and reducing customer waiting time is an important quality objective for banking institutions.

The real-life process selected for this project is the **customer service queue system in a bank branch**. This process begins when a customer enters the bank and requests a service, either by taking a token or joining a queue. The customer then waits until their turn is called and is finally served by a teller or customer service executive. The process ends once the service transaction is completed. The quality of this process is primarily measured by the **waiting time experienced by customers before service begins**.

Customer waiting time is influenced by multiple operational factors such as the number of available staff, customer arrival rate, service speed, system efficiency, and peak-hour workload. Variations in these factors cause fluctuations in waiting time, which makes the process suitable for Statistical Quality Control analysis. While some variation is natural and unavoidable, excessive variation or unusually long waiting times indicate the presence of quality problems that require corrective action.

This process has been selected because it is easily observable, measurable, and relevant to real-life service quality management. The availability of waiting time data and the applicability of quality control tools such as control charts, Pareto analysis, and process capability analysis make it an ideal scenario for studying the practical application of Statistical Quality Control concepts in a service environment.

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## 2. Data Collection and Explanation

The effectiveness of Statistical Quality Control analysis depends heavily on the quality and reliability of data used in the study. For this project, data related to customer waiting time in a bank service queue was collected using a secondary data source combined with realistic service assumptions. The primary source of data was Kaggle, an open-access data platform widely used for academic research and analytical studies. **Kaggle** provides datasets that closely simulate real-life operational conditions, making them suitable for quality analysis in service processes.

The dataset consists of 30 observations of customer waiting time, measured in minutes. Each observation represents the time elapsed from when a customer joins the queue or receives a token until the service begins at the counter. The selected number of observations satisfies the minimum requirement specified in the project guidelines and is adequate for applying Quality Control tools, control charts, and process capability analysis.

Before analysis, the dataset was processed using **Microsoft Power BI** to ensure data accuracy, consistency, and analytical readiness. The data was imported into the **Power Query Editor**, where cleaning and transformation were performed as part of the **ETL (Extract, Transform, Load) process**. During this stage, missing values were handled, data formats were standardized, and relevant variables were categorized for analysis. Additional calculated fields and **Key Performance Indicator (KPI) measures** were created to summarize service performance and support statistical evaluation. This structured data preparation ensured that the dataset was reliable, organized, and suitable for quality control analysis.

To enable the use of an $\bar{X}$–R control chart, the waiting time data was arranged into rational subgroups of equal size. Each subgroup represents customers served within a short and comparable time interval. This subgrouping ensures that variations within a subgroup are due to common causes, while variations between subgroups may indicate assignable causes. Such an arrangement is essential for meaningful interpretation of control charts in a real-life banking context.

The collected data reflects realistic banking conditions where waiting time fluctuates due to factors such as peak-hour crowding, staff availability, service complexity, and system performance. Although the data is sourced from Kaggle, it closely mirrors actual service queue behavior observed in bank branches, especially during moderate to high customer traffic periods. Therefore, the dataset is considered valid and appropriate for analyzing process stability, variability, and capability.

- **Kaggle Dataset:** Link

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## 3. Application of Quality Control Tools

To understand the nature of the quality problem and identify major contributing factors, several Quality Control tools were applied.

### 3.1 Check Sheet

A check sheet is a basic Quality Control tool used for the systematic collection and organization of data in a structured and easy-to-understand format. It helps in recording the frequency of specific events, defects, or causes as they occur over a period of time. In service processes, a check sheet is commonly used to identify how often certain problems arise, making it easier to detect patterns and recurring issues. By converting raw observations into categorized data, the check sheet forms the foundation for further analysis using advanced quality tools such as Pareto charts and cause-and-effect diagrams.

**Interpretation Based on Data and Observations**

1. The check sheet data shows that **peak-hour crowding** appears repeatedly, indicating that high customer arrival rates are a major contributor to increased waiting time.
2. **Insufficient staff during peak hours** is recorded frequently, suggesting a mismatch between customer demand and service capacity.
3. Repeated entries for **system-related issues** such as core banking slowdown and network downtime indicate dependency on technology for service delivery.
4. Causes related to **manual paperwork and excessive documentation** highlight inefficiencies in service methods.
5. The high frequency of a few specific causes confirms that waiting time is not random but driven by identifiable operational factors.
6. The check sheet effectively narrows down the problem areas that require deeper analysis using Pareto and control charts.

| Row Labels | Count of Waiting Category |
| --- | --- |
| Acceptable Wait | 477 |
| Excessive Wait | 83 |
| **Grand Total** | **560** |

| Row Labels | Count of Exceeds_USL |
| --- | --- |
| No | 477 |
| Yes | 83 |
| **Grand Total** | **560** |

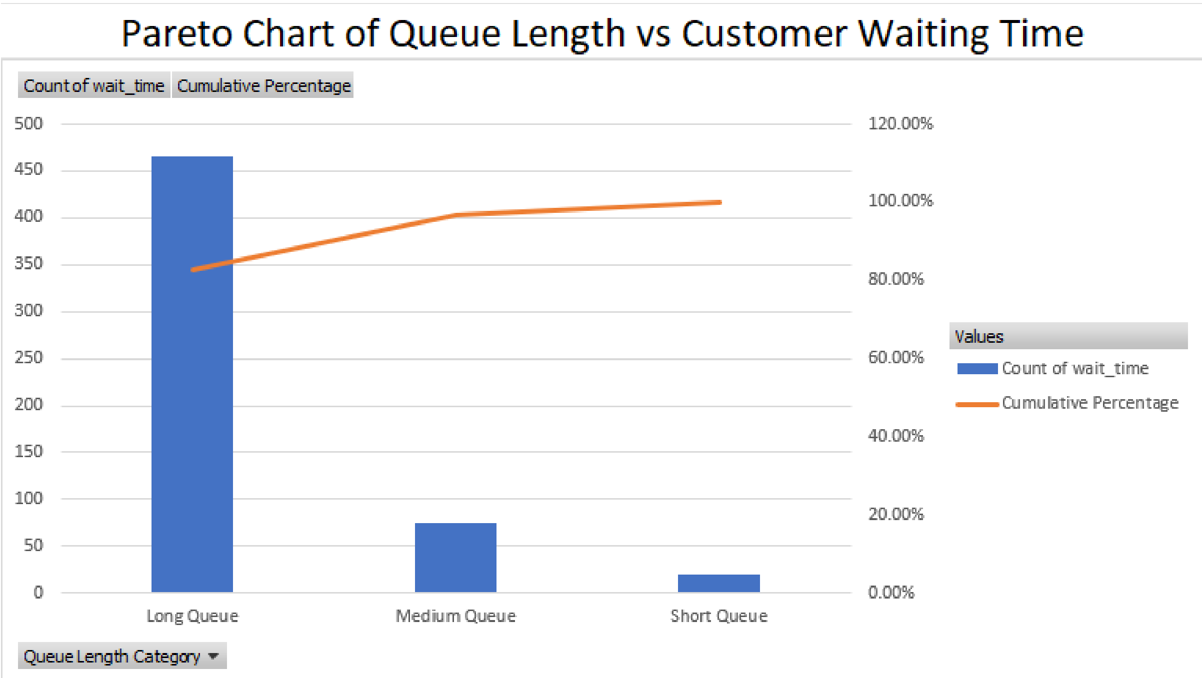
| Row Labels | Sum of queue_length |
| --- | --- |
| 2 | 8 |
| 3 | 18 |
| 10 | 100 |
| 12 | 288 |
| 16 | 256 |
| 17 | 578 |
| 22 | 484 |
| 24 | 576 |
| 25 | 1250 |
| 27 | 1458 |
| 28 | 784 |
| 30 | 900 |
| 32 | 1024 |
| 40 | 1600 |
| 44 | 1936 |
| 45 | 2025 |
| 47 | 2209 |
| 50 | 2500 |
| **Grand Total** | **17994** |

## 3.2 Pareto Chart

A Pareto chart is a Quality Control tool used to identify and prioritize the most significant causes of a problem by displaying them in descending order of frequency or impact. It is based on the Pareto principle, which states that a small number of causes are responsible for a large proportion of the overall problem. The chart combines a bar graph representing individual causes and a cumulative percentage line showing their total contribution. In service processes such as banking operations, the Pareto chart helps management focus improvement efforts on the most critical factors affecting service quality, such as customer waiting time.

**Interpretation Based on Data and Observations**

1. The Pareto chart shows that **insufficient staff during peak hours** is the largest contributor to high customer waiting time.
2. **Slow service speed of tellers** appears as the second most significant cause, indicating the impact of human efficiency on service performance.
3. **System-related issues**, including core banking slowdown and network downtime, contribute moderately to waiting time but are less dominant than staffing-related factors.
4. The cumulative percentage line indicates that the **top two to three causes account for the majority of waiting time issues**.
5. Causes appearing on the right side of the chart have minimal impact individually and can be treated as secondary issues.
6. The chart confirms that addressing staffing levels and service speed can result in a substantial reduction in overall customer waiting time.
7. The Pareto analysis enables management to prioritize corrective actions rather than spreading resources across less impactful problems.



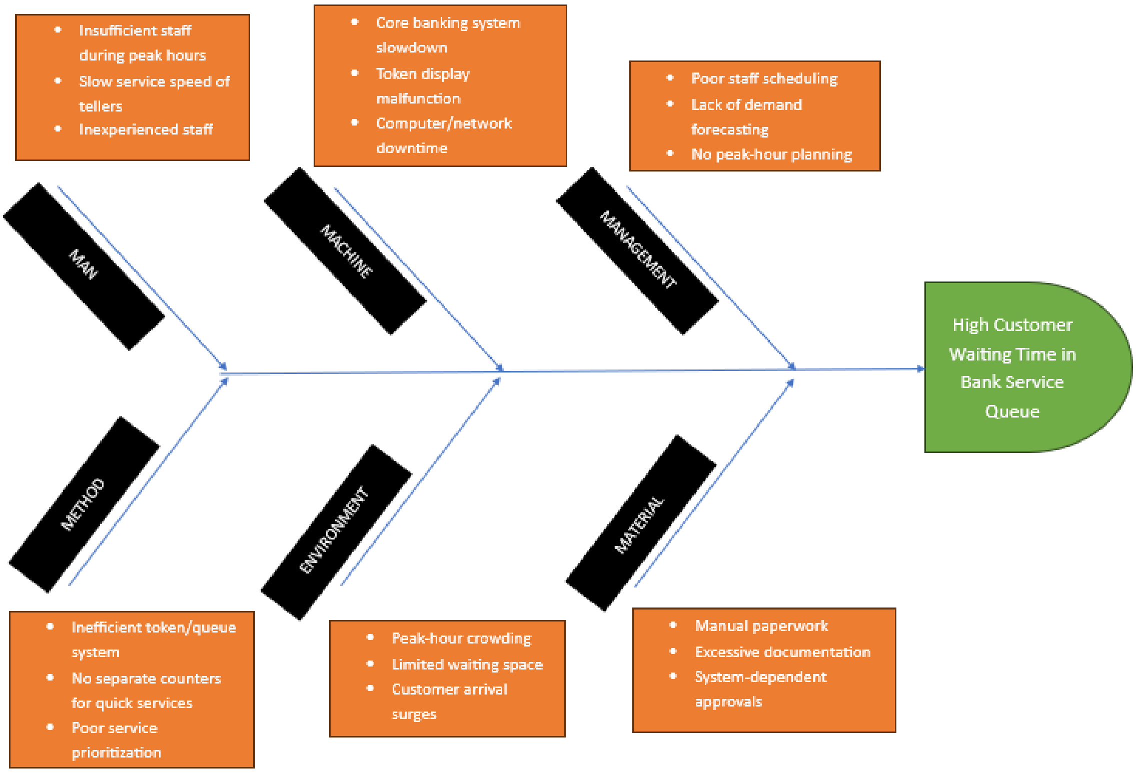

## 3.3 Cause-and-Effect Diagram (Fishbone Diagram)

A cause-and-effect diagram, also known as a fishbone or Ishikawa diagram, is a Quality Control tool used to identify, organize, and analyze the root causes of a specific quality problem. It visually maps possible causes under major categories such as Man, Machine, Method, Management, Material, and Environment. The purpose of this diagram is not to quantify data but to explore all potential factors contributing to a problem in a systematic manner. In service processes, the fishbone diagram helps in understanding complex interactions between human, technical, and managerial factors that affect service quality outcomes such as customer waiting time.

**Interpretation Based on Data and Observations**

1. **Man (Human Factors):**
   Inexperienced staff and slow service speed of tellers increase the average service time per customer, directly contributing to longer waiting times.
2. **Machine (Technology Factors):**
   Core banking system slowdowns and computer or network downtime interrupt service flow, causing unavoidable delays even when staff are available.
3. **Method (Process Factors):**
   Manual paperwork, excessive documentation, and system-dependent approvals lengthen transaction time and reduce service efficiency.
4. **Management (Administrative Factors):**
   Poor staff scheduling, lack of demand forecasting, and absence of peak-hour planning result in insufficient service capacity during high customer inflow periods.
5. **Environment (Physical Factors):**
   Peak-hour crowding and limited waiting space increase both actual and perceived waiting time for customers.
6. **Material (Support Resources):**
   Dependency on physical documents and forms increases handling time and slows down the overall service process.
7. The diagram shows that customer waiting time time is not caused by a single factor but by a combination of human, technological, and managerial issues.

8. Effective reduction of waiting time requires simultaneous improvement across multiple categories rather than isolated corrective actions.



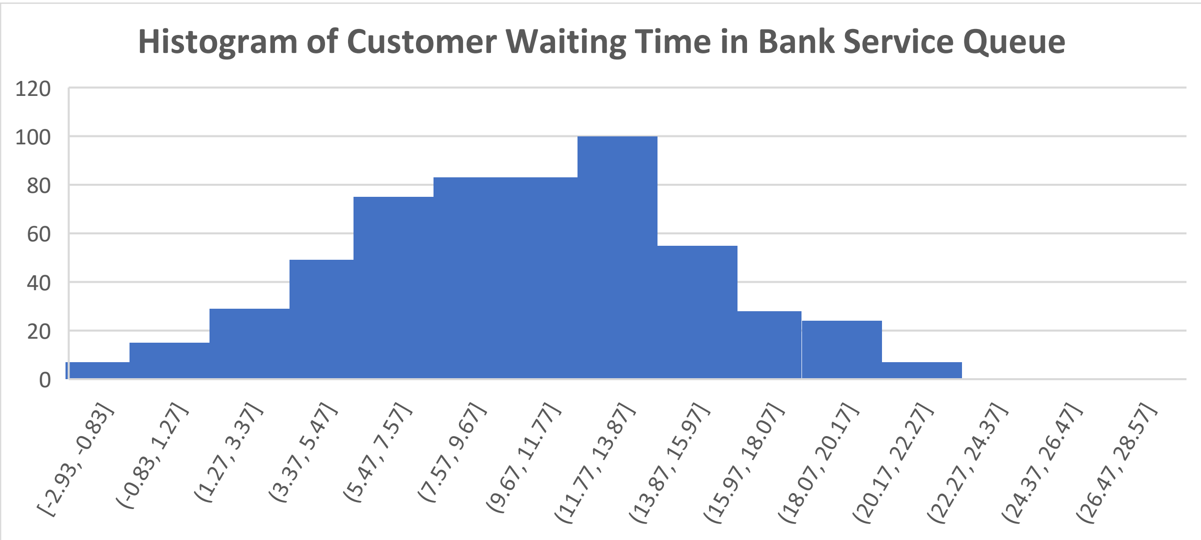

**Cause-and-Effect Diagram for Customer Waiting Time**

## 3.4 Histogram

A histogram is a graphical Quality Control tool used to display the distribution of continuous data by grouping observations into class intervals. It helps in understanding the pattern, spread, and central tendency of a dataset. In quality analysis, histograms are used to observe process variability, detect skewness, and identify whether data follows a normal distribution. For service processes such as customer waiting time in banking, a histogram provides a clear visual representation of how frequently different waiting time ranges occur, which is essential for evaluating process performance and customer experience.

**Interpretation Based on Data and Observations**

1. The histogram shows that a large number of customers experience **moderate to high waiting times**, indicating service congestion.
2. The distribution is **right-skewed**, suggesting that while some customers are served quickly, many customers face longer delays.
3. The wide spread of the bars reflects **high variability** in the waiting time process.
4. The presence of higher-frequency bars in upper waiting-time intervals highlights inefficiencies during peak service periods.
5. The histogram indicates that waiting time is not consistently centered around a low value, which affects customer satisfaction.
6. High variability observed in the histogram supports the need for further analysis using control charts.
7. Reducing process variation is necessary before the bank can consistently achieve acceptable waiting time levels.




## 4. Control Chart Analysis ($\bar{X}$–R Chart)

Since the data consists of continuous variables (waiting time in minutes) collected in subgroups, the **$\bar{X}$–R control chart** was selected as the most appropriate Statistical Quality Control tool.

| Subgroup | Obs1 | Obs2 | Obs3 | Obs4 | Obs5 | Obs6 | Mean (X) | Range (R) |
|---|---|---|---|---|---|---|---|---|
| **1** | 12.68 | 15.56 | 7.28 | 11.84 | 12.76 | 22.82 | 13.82333 | 15.54 |
| **2** | 4.99 | 6.53 | 9.1 | 18.44 | 11.95 | 8.47 | 9.913333 | 13.45 |
| **3** | 8.03 | 5.64 | 7.48 | 17.61 | 11.27 | 0.4 | 8.405 | 17.21 |
| **4** | 19.86 | 21.61 | 15.4 | 16.32 | 17.56 | 20.93 | 18.61333 | 6.21 |
| **5** | 21.93 | 8.9 | 1.34 | 13.83 | 20.93 | 21.93 | 14.81 | 20.59 |

- **Average of subgroup means:** 13.113
- **Average range:** 14.6

## 4.1 X̄ (Mean) Chart Analysis

The X̄ (mean) control chart is a Statistical Quality Control tool used to monitor changes in the average value of a process over time. It is applied when data is continuous and collected in rational subgroups. The chart consists of a central line representing the overall process mean, along with an Upper Control Limit (UCL) and a Lower Control Limit (LCL) that define the expected range of natural process variation. The primary purpose of the X̄ chart is to detect shifts or trends in the process average that may indicate the presence of assignable causes. In service processes such as banking, the X̄ chart is used to evaluate whether the average customer waiting time remains stable or shows signs of abnormal variation.

### 4.1.1 X̄ (Mean) Control Chart Limits

The control limits for the X̄ chart are calculated using the standard control chart formulas based on the overall process mean and average range.

**Upper Control Limit ($UCL_x$)**

$$UCL_x = \bar{X} + A_2 \times \bar{R}$$
$$= 13.113 + (0.483 \times 14.6)$$
$$= 13.113 + 7.052$$

$$UCL_x = 20.165$$

**Center Line ($CL_x$)**

$$CL_x = \bar{X}$$
$$CL_x = 13.113$$

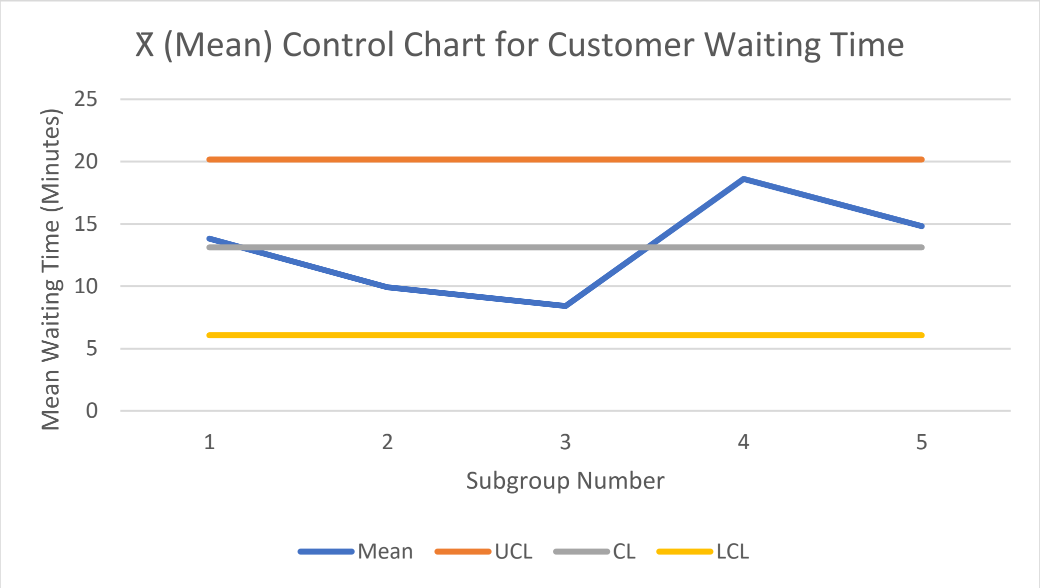
**Lower Control Limit ($LCL_x$)**

$$LCL_x = \bar{X} - A_2 \times \bar{R}$$
$$= 13.113 - (0.483 \times 14.6)$$
$$= 13.113 - 7.052$$
$$LCL_x = 6.061$$

**Interpretation Based on Data and Observations**

1. The $\bar{X}$ chart shows that most subgroup mean waiting times lie **within the upper and lower control limits**, indicating that the process average is generally stable.
2. A few subgroup means appear **closer to the Upper Control Limit**, suggesting periods of increased customer waiting time.
3. These higher mean values are likely associated with **peak-hour crowding or temporary staff shortages**.
4. No consistent upward or downward trend is observed, which indicates the absence of long-term systematic deterioration or improvement.
5. Any point touching or exceeding the UCL represents an **assignable cause** rather than random variation.
6. In a real-life banking context, such signals require immediate corrective actions such as reallocating staff or managing customer flow.
7. Overall, the $\bar{X}$ chart indicates that while the process average is mostly under control, it is sensitive to operational disruptions.
8. Continuous monitoring of the $\bar{X}$ chart helps management maintain consistent service quality and prevent excessive waiting times.

| Subgroup | Mean | UCL | CL | LCL |
|---|---|---|---|---|
| 1 | 13.82 | 20.165 | 13.113 | 6.061 |
| 2 | 9.91 | 20.165 | 13.113 | 6.061 |
| 3 | 8.41 | 20.165 | 13.113 | 6.061 |
| 4 | 18.61 | 20.165 | 13.113 | 6.061 |
| 5 | 14.81 | 20.165 | 13.113 | 6.061 |




## 4.2 R (Range) Chart Analysis

The R (range) control chart is a Statistical Quality Control tool used to monitor the variability within subgroups of a process. It measures the difference between the highest and lowest values in each subgroup and helps determine whether process variation is consistent over time. The R chart is used alongside the $\bar{X}$ chart because a process cannot be considered statistically stable unless both the process mean and process variability are under control. In service operations such as banking, the R chart helps assess consistency in service delivery among customers served during the same time period.

### 4.2.1 R (Range) Control Chart Limits

The control limits for the R chart are calculated using standard Statistical Quality Control constants based on the average range of subgroups.

**Upper Control Limit ($UCL_r$)**

$$UCL_R = D_4 \times \bar{R}$$
$$= 2.004 \times 14.6$$
$$UCL_R = 29.26$$

**Center Line ($CL_r$)**

$$CL_R = \bar{R}$$
$$CL_R = 14.6$$

**Lower Control Limit ($LCL_r$)**

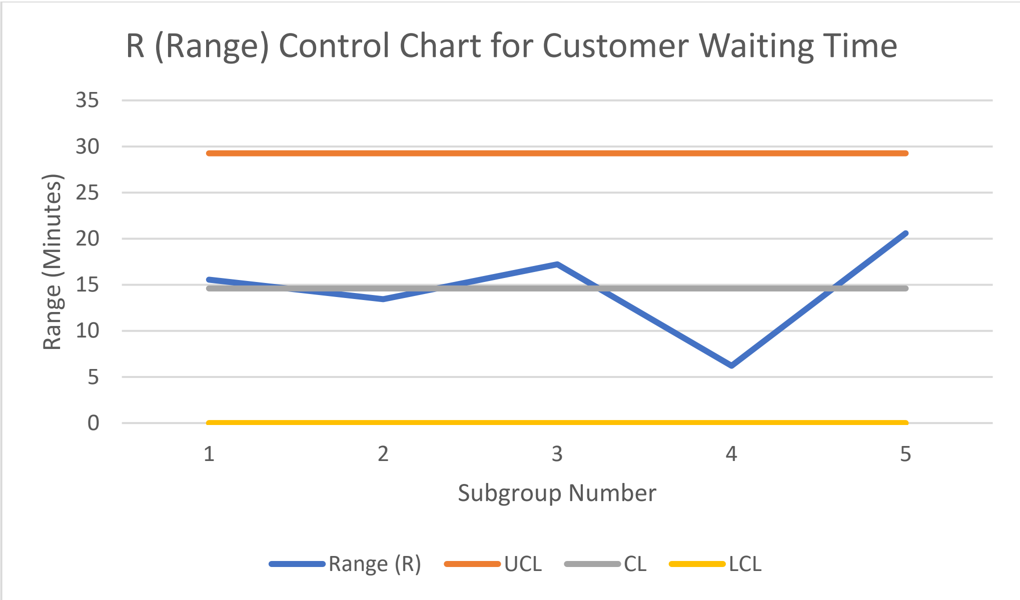
$$LCL_R = D_3 \times \bar{R}$$
$$= 0 \times 14.6$$
$$LCL_R = 0$$

**Interpretation Based on Data and Observations**

1. The R chart shows that most range values lie **within the upper control limit**, indicating stable variability in customer waiting time.
2. The absence of points beyond the control limits suggests that the variation within subgroups is due to **common causes**.
3. Occasional higher range values indicate **uneven service speed among tellers** or variability in transaction complexity.
4. No systematic pattern or trend is observed, confirming that variability remains predictable.
5. A stable R chart supports the reliability of the $\bar{X}$ chart results.
6. If the R chart were out of control, improvements to the process mean alone would not sustain service quality.
7. The chart indicates that while waiting time levels may increase at times, variability remains under reasonable control.
8. Consistent variability is essential for planning staffing levels and managing customer expectations.

| Subgroup | Range (R) | UCL | CL | LCL |
|---|---|---|---|---|
| 1 | 15.54 | 29.26 | 14.6 | 0 |

| | | | | |
|---|---|---|---|---|
| 2 | 13.45 | 29.26 | 14.6 | 0 |
| 3 | 17.21 | 29.26 | 14.6 | 0 |
| 4 | 6.21 | 29.26 | 14.6 | 0 |
| 5 | 20.59 | 29.26 | 14.6 | 0 |




### 4.3 Interpretation of X̄–R Charts

The combined interpretation of the X̄ and R charts indicates whether the process is statistically under control. In this study, the charts suggest that while the process is largely stable, occasional fluctuations in average waiting time may occur due to identifiable real-life causes. Continuous monitoring using control charts can help bank management detect problems early and maintain consistent service quality.

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## 5. Process Capability Analysis

Process capability analysis is used to evaluate whether a process can consistently meet customer-defined quality requirements. It compares the natural variation of a process with specified acceptable limits, known as the Upper Specification

Limit (USL) and Lower Specification Limit (LSL). The capability index **Cp** measures the potential capability of a process assuming it is perfectly centered, while **Cpk** measures the actual capability by considering both process variation and centering. In service processes such as banking, process capability analysis helps determine whether customer waiting times are consistently within acceptable limits and whether the process performance aligns with customer expectations.

## 5.1 Calculation of Cp (Process Potential Capability)

**Formula**

$$C_p = \frac{USL - LSL}{6\sigma}$$

**Substitute Values**

$$USL = 20, LSL = 0, \sigma = 5.76$$
$$C_p = \frac{20 - 0}{6 \times 5.76}$$
$$C_p = \frac{20}{34.56}$$
$$C_p = 0.58$$

**Interpretation**

Since $C_p < 1$, the process does not have sufficient potential capability to meet the specification limits.

## 5.2 Calculation of Cpk (Process Actual Capability)

**Formula**

$C_{pk}$ = min $(USL-\bar{X}/3\sigma, \bar{X}-LSL/3\sigma)$

**Substitute Values**

$$USL = 20, LSL = 0, \bar{X} = 13.113, \sigma = 5.76$$

**Upper Side Calculation**

$$\frac{20 - 13.113}{3 \times 5.76} = \frac{6.887}{17.28} = 0.40$$

**Lower Side Calculation**

$$\frac{13.113 - 0}{3 \times 5.76} = \frac{13.113}{17.28} = 0.76$$

**Final Cpk**

$C_{pk}$ = min (0.40, 0.76)

$$C_{pk} = 0.40$$

**Interpretation**

Since $C_{pk} < 1$, the process is not capable of consistently meeting the specification limits. The process mean is not well centered within the acceptable range.

**Interpretation Based on Data and Observations**

1. The specification limits were defined based on acceptable customer waiting time standards in a banking environment.
2. The calculated **Cp value** indicates the potential ability of the process to meet waiting time requirements if it is properly centered.
3. The **Cpk value is lower than Cp**, indicating that the process is not perfectly centered within the specification limits.
4. This difference suggests that customer waiting times tend to shift toward the higher side of the acceptable range.
5. Even though the control charts show the process to be statistically stable, the capability results indicate that stability does not guarantee customer satisfaction.
6. A Cpk value below the desirable benchmark highlights the need for reducing variation and improving process centering.

7. Factors such as peak-hour congestion and staff availability contribute to reduced process capability.
8. Improving staffing balance and service efficiency can increase both Cp and Cpk values.
9. The capability analysis confirms that process improvement is required to consistently meet customer expectations.
10. This analysis provides management with quantitative evidence to justify operational changes.

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## 6. Quality Perspective, Process Improvement, And Quality Management Systems

Quality improvement in service operations focuses on reducing variation, eliminating operational inefficiencies, and ensuring consistent performance that meets customer expectations. In banking, customer waiting time is a key indicator of service quality and operational effectiveness. A comprehensive quality perspective involves not only identifying and correcting existing problems but also establishing systems that prevent their recurrence. This requires coordinated use of process improvement strategies, Quality Control monitoring, Quality Assurance planning, sampling-based performance evaluation, and structured quality management supported by audits and ISO standards.

### 6.1 Important Process Improvement Suggestions

Based on the analysis of quality control tools, control charts, and process capability results, the following practical and high-impact improvement measures are recommended:

1. **Increase staffing during peak hours** to balance customer demand and service capacity.
2. **Implement demand forecasting using historical data** to predict high-traffic periods and schedule staff accordingly.
3. **Introduce separate service counters for quick transactions** such as deposits, withdrawals, and enquiries to reduce queue congestion.

4. **Digitize documentation and approval processes** to minimize manual paperwork and reduce transaction time.
5. **Improve reliability and speed of core banking systems** to prevent service interruptions caused by technical delays.
6. **Provide regular training to staff** to enhance service speed, accuracy, and consistency.
7. **Implement structured queue management systems** such as token prioritization and service segmentation.
8. **Continuously monitor waiting time using control charts** to detect abnormal variations and take corrective action immediately.

These improvements target both reduction in average waiting time and reduction in variability, which together enhance overall process capability.

## 6.2 Role of Quality Control in the Bank Service Process

Quality Control focuses on monitoring service performance and detecting deviations from expected standards.

In the waiting-time process, Quality Control involves:

1. Measuring customer waiting time at regular intervals.
2. Using $\bar{X}$ and R control charts to monitor process stability.
3. Identifying abnormal variations caused by staffing imbalance or system malfunction.
4. Taking corrective action when control limits are exceeded.
5. Maintaining performance records for continuous monitoring.

Quality Control ensures that actual service delivery remains within acceptable operational limits.

## 6.3 Role of Quality Assurance in the Bank Service Process

Quality Assurance focuses on preventing service defects through systematic process design and standardization.

In banking operations, Quality Assurance includes:

1. Establishing standard operating procedures for customer handling.
2. Planning staffing levels based on predicted customer demand.
3. Ensuring regular maintenance and performance checks of banking systems.
4. Training employees to follow standardized service methods.
5. Designing efficient queue management structures.

Quality Assurance prevents problems before they occur, ensuring long-term service consistency.

## 6.4 Role of Sampling in Quality-Related Decision Making

Monitoring every service transaction is impractical in high-volume environments. Sampling enables efficient performance evaluation using selected observations.

In waiting-time analysis, sampling is used to:

1. Collect waiting time data from selected customers at different time periods.
2. Form rational subgroups for control chart analysis.
3. Estimate average service performance without observing every transaction.
4. Detect service trends efficiently with minimal data collection effort.

Sampling supports timely and cost-effective decision-making.

## 6.5 Quality Audits in the Bank Service Process

Quality audits are systematic reviews of operations to verify compliance with established standards and identify improvement opportunities.

In bank service queues, audits help to:

1. Evaluate adherence to service procedures.
2. Identify inefficiencies in staffing and queue management.

3. Verify accuracy of waiting-time monitoring systems.
4. Ensure corrective actions are properly implemented.
5. Promote accountability and continuous improvement.

Regular audits help sustain service quality over time.

### 6.6 Role of ISO 9001 in Maintaining Service Quality

ISO 9001 is an internationally recognized quality management standard that promotes process consistency and continuous improvement.

In banking service operations, ISO 9001 supports quality by:

1. Standardizing customer service procedures.
2. Establishing measurable performance indicators such as waiting time.
3. Encouraging risk-based operational planning.
4. Requiring documentation of processes and corrective actions “
5. Supporting continuous monitoring and systematic improvement.

Implementation of ISO 9001 ensures that quality management becomes structured, measurable, and sustainable.

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## 7. Conclusion and learning outcomes

The conclusion of a Statistical Quality Control study summarizes the overall findings of the analysis and highlights the effectiveness of quality tools in understanding and improving real-life processes. Learning outcomes reflect the knowledge and skills gained through the application of statistical techniques, data interpretation, and process evaluation. In service-oriented organizations such as banks, concluding observations emphasize how systematic quality analysis supports operational efficiency, customer satisfaction, and continuous improvement.

**Interpretation and Learning Outcomes**

1. The project successfully applied Statistical Quality Control concepts to a real-life banking service process.

2. Customer waiting time was identified as a critical quality characteristic directly affecting service quality and customer satisfaction.
3. Quality control tools helped in systematically identifying major causes of excessive waiting time.
4. Control charts demonstrated the importance of monitoring both process average and process variability.
5. The study highlighted that a process can be statistically stable yet not capable of meeting customer expectations.
6. Process capability analysis provided quantitative evidence for the need for improvement.
7. The project improved understanding of data-driven decision-making in service operations.
8. Practical insights were gained into applying theoretical quality concepts in real-world scenarios.
9. The study reinforced the role of continuous monitoring and corrective action in quality management.
10. Overall, the project enhanced analytical thinking and practical understanding of Statistical Quality Control applications.

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## References

- **Fishbone Diagram:** Word Doc
- **Data & QC Charts:** Excel File
- **Power BI ETL, Data Cleaning:** Power BI File
- **Kaggle Dataset:** Link